Amr Salah Shaheen

Machine Learning Engineer

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## PROFILE SUMMARY

- Third-year Computer Science and Artificial Intelligence student at Helwan University.
- Machine Learning Engineer with practical experience in building and deploying end-to-end ML projects, including regression, classification, clustering, and association models.
- Skilled in data cleaning, preprocessing, exploratory data analysis (EDA), and feature engineering, with strong proficiency in Python and model optimization to deliver data-driven solutions.

## TECHNICAL SKILLS

**Machine Learning:** Supervised Learning, Unsupervised Learning, Model Evaluation & Hyperparameter Tuning
**Data Analysis:** EDA, Data Cleaning, Preprocessing, Feature Engineering
**Programming Languages:** Python, Java, C, C++
**Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Streamlit
**Tools:** SQL, Git

## PROJECTS

### Diamond Price Predictor | GitHub

- ML regression pipeline to estimate diamond prices from quality grades and physical measurements.
- Applied data cleaning, preprocessing, feature engineering to reduce multicollinearity, and EDA.
- Trained and evaluated regression models using 5-fold cross-validation, tuned top candidates, and selected the final model achieving $R^2 > 0.98$; Deployed an interactive Streamlit web application for real-time price prediction.

### Telecom Churn Predictor | GitHub

- ML classification pipeline to predict telecom customer churn from customer demographics, account details, and service usage patterns.
- Applied data cleaning, preprocessing, EDA, feature engineering, and handled class imbalance.
- Trained and evaluated classification models using stratified cross-validation, tuned top candidates, selected the best model achieving Recall = 0.89 and ROC-AUC = 0.84, optimized decision threshold (0.40) to improve churn detection for retention targeting, and deployed a Streamlit web application for real-time churn prediction.

### Spam/Ham Classifier | GitHub

- Binary text classification system for spam detection in SMS and email messages.
- Applied message cleaning, preprocessing, tokenization, and compared stemming vs. lemmatization techniques.
- Evaluated multiple vectorization strategies, benchmarked and tuned Naive Bayes variants, selected the best model, and deployed a Streamlit web app for real-time message classification.

### Online Retail Customer Segmentation | GitHub

- Customer segmentation pipeline to group retail customers based on purchasing behavior and transaction patterns
- Applied data cleaning, preprocessing, customer-level feature engineering, and EDA.
- Compared K-Means and DBSCAN models, performed clustering evaluation, selected K-Means as the final model, applied business-driven cluster profiling, and deployed a Streamlit web app for customer segment prediction.

### Association-Based Movie Recommendations | GitHub

- Developed an association-rule-based recommendation system on MovieLens dataset (9,742 movies, 100,836 ratings, 610 users) by modeling each user as a transaction of positively rated items.
- Performed EDA and data preprocessing, including rating filtering (≥ 3.5), user-level basket construction, noise reduction, and one-hot encoding; identified basket size insights.
- Implemented and compared Apriori and ECLAT with tuned support, confidence, and lift, generating 29,227 itemsets and 62,937 rules; ECLAT outperformed in time and memory, selected as the final model and export for deployment.

## SCHOLARSHIP & TRAINING

**Digital Egypt Pioneers Initiative (DEPI)** Dec 2025 – Present
AI & Machine Learning Trainee

- AI and Machine Learning scholarship program focused on applied Data Analysis, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and model development.

## EDUCATION

**Helwan University** Oct 2023 – Present
B.Sc. Computer Science and Artificial Intelligence

## CERTIFICATIONS

- **AI and Machine Learning** — Sprints x Microsoft Summer Camp
- **Mathematics for Machine Learning and Data Science** — DeepLearning.AI - Coursera